

Full-Stack Developer building production SaaS platforms, cross-platform mobile ecosystems, and applied AI systems end to end — from database schema and API design to deployed, production-grade UI. Experience spans multi-vendor commerce, institutional ERP, mobile AR, and computer-vision pipelines using Vision Transformers. Seeking full-time Full-Stack, Web, and Mobile Development roles.

SKILLS

Domains	Full-Stack Web & Mobile Development, Applied AI & Computer Vision, SaaS Architecture, Augmented Reality / Spatial Computing
Languages	TypeScript, JavaScript, Python, Java
Frameworks/Libraries	React 19, Next.js 16, React Native (Expo SDK 54), Node.js, Express.js, PyTorch (ViT), FastAPI, Three.js, Tailwind CSS v4
Tools/Platforms	Git, GitHub, Firebase Suite, MongoDB, PostgreSQL, REST APIs

EXPERIENCE

Web Development Intern Aug 2025 – Oct 2025	ARCH TECHNOLOGIES Remote, Pakistan
---	---------------------------------------

- Worked as a Web Development Intern, contributing to front-end and back-end feature development on live projects.
- Built responsive frontend UI and API endpoints using React, Next.js, and Node.js; diagnosed and resolved production issues while optimizing load performance.

Full-Stack Developer 2024 – Present	INDEPENDENT / FREELANCE Remote
--	-----------------------------------

- Designed and shipped 6+ end-to-end web, mobile, and applied-AI systems spanning SaaS, e-commerce, ERP, and computer vision.
- Engineered production database schemas and REST APIs using Next.js, Node.js, PostgreSQL, and MongoDB.
- Fine-tuned Vision Transformer (ViT) and CNN models and deployed FastAPI inference microservices with sub-second latency.
- Built cross-platform mobile applications in React Native/Expo integrating AR (Viro React), Firebase, and Stripe payments.

ACHIEVEMENTS

Digital Pakistan Speed Programming Competition (DPSPC) National Speed Programming & Algorithmic Problem Solving Competition	TOP 50 NATIONWIDE 2025
--	---------------------------

- Ranked in the Top 50 teams nationwide out of ~450 competing teams, tackling speed programming challenges evaluating problem solving, data structures, and optimal time/space complexity under strict limits.

InnovateX 2.0 Coding Competition Senior-Level Competitive Programming & Development Hackathon by Brackets	GIFT UNIVERSITY GIFT University
--	------------------------------------

- Competed in a senior-level algorithmic coding competition focused on data structures, problem solving, and time-complexity optimization under strict contest deadlines.
- Solved competitive programming challenges emphasizing efficient algorithm design and optimal time/space complexity, recognized among top performers by industry evaluators.

PROJECTS

GardenMate Ecosystem Mobile App & Computer Vision API (React Native, Expo, PyTorch ViT-B/16, Viro AR, FastAPI, Firebase)	FINAL YEAR PROJECT 2025 – 2026
---	-----------------------------------

- AI-powered, AR-integrated smart gardening assistant and plant marketplace with 12+ specialized modules across four role-based portals.
- Fine-tuned a Vision Transformer (ViT-B/16) classifier for 69 plant species, fused with the PlantNet API in a 60/40 weighted ensemble, reaching ~350ms inference latency.
- Implemented Viro React AR plane detection for a "Place Before You Buy" 3D plant-placement experience.
- Built a Snap-to-Autolist pipeline letting vendors auto-populate product listings from a single photo.
- Designed a 10% commission engine with Stripe settlement across Customer, Vendor, Rider, and Admin portals.

Vouch Keychain Full-Stack SaaS Platform (Next.js 16, React 19, Firebase Suite, Twilio, Resend, Three.js)	SAAS PLATFORM 2026
---	-----------------------

- Smart QR-coded keychains reuniting lost items with owners via privacy-preserving, masked-contact communication.
- Built real-time location-recovery telemetry, in-app messaging, and an interactive Three.js 3D product visualizer.

PROJECTS (CONTINUED)

NutriLens AI <i>Mobile Application (React Native, Expo SDK 54, Zustand 5, Firebase Realtime Database)</i> • Multimodal vision-nutrition engine converting food photography and grocery barcodes into real-time macronutrient telemetry. • Implemented clinical Mifflin-St Jeor metabolic equations and an AI dietitian coach powered by OpenRouter AI.	SMART NUTRITION APP 2025
Evershine Academy ERP & LMS <i>Full-Stack Web Platform (Next.js 16, React 19, TypeScript, PostgreSQL)</i> • Multi-campus educational ERP/LMS with on-premise biometric fingerprint machine sync for staff and student attendance. • Built 7 role-based portals, automated fee-challan generation, and staff payroll ledgers.	INSTITUTIONAL SYSTEM 2026
FinGo Marketplace <i>Full-Stack E-Commerce Platform (React.js, Node.js, Express.js, MongoDB, JWT)</i> • Multi-vendor marketplace connecting retail and grocery stores with customers across Lahore, Karachi, and Islamabad. • Implemented 3-tier role-based access control, JWT authentication with refresh-token rotation, and real-time inventory tracking.	MULTI-VENDOR COMMERCE 2024
Voice AI Copilot <i>Web Application (Groq LLaMA 3.3 70B, Groq Whisper Large v3, Web Audio API)</i> • Browser-based voice assistant converting spoken commands into structured task and reminder actions via LLM tool calling. • Integrated Groq LLaMA 3.3 70B and Groq Whisper Large v3 with the Web Audio API for real-time speech processing.	VOICE ASSISTANT 2025 – 2026

EDUCATION

Bachelor of Science in Computer Science (BS CS) <i>Coursework: Software Engineering, Data Structures & Algorithms, Database Systems, Artificial Intelligence, Machine Learning, Computer Vision, Distributed Systems</i>	GIFT UNIVERSITY, GUJRANWALA 2022 – 2026
Intermediate in Pre-Engineering (FSc) <i>Higher Secondary School Certificate</i>	GOVT COLLEGE 2020 – 2022

LINKS

Portfolio:	umairulhassan.site
GitHub:	github.com/umairulhassancs
LinkedIn:	linkedin.com/in/umair-ul-hassan-cs